AnchorMTM: Spatially Anchored Merge Tree Maps for Time-Varying Scalar Fields

Category: Research

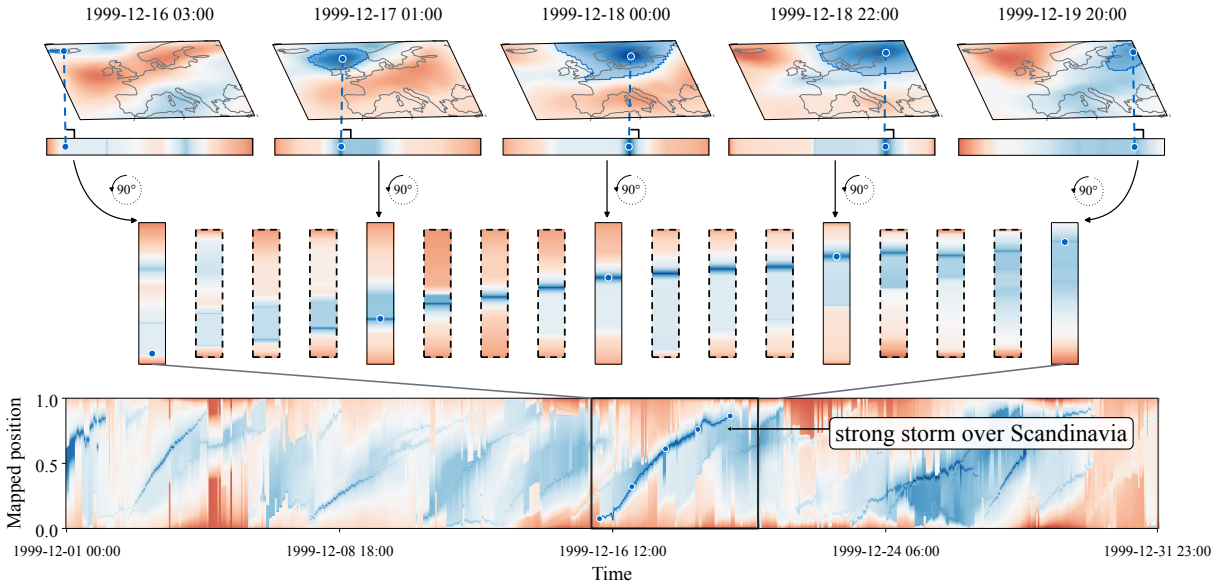


Figure 1: Storms example linking spatial feature supports to a temporal map ($\varepsilon = 5\%$). Top: five snapshots of one matched extremum and their scalar profiles; blue outlines mark its leaf-arc supports, and dashed guides link the extrema to their displayed profile anchors. Middle: the profiles rotated by 90°; dashed borders distinguish sampled intermediate frames. Bottom: the full 744-frame map, with the selected storm-related region outlined and the matched feature's displayed positions marked in blue.

Abstract

Temporal merge tree maps summarize evolving scalar topology, but their display coordinates need not correspond to a common spatial direction. AnchorMTM links these coordinates to a fixed reference estimated once from matched extremum motion across a 2D or 3D sequence. The leading eigenvector of the motion scatter matrix maximizes squared projected motion; a physically meaningful direction can alternatively be supplied. This standard estimation step provides targets for topology-budgeted layout rather than a separate direction-optimization contribution. A finite heuristic search seeks a leaf order with small worst reference displacement under a declared merge-level error budget. Exact fixed-order displacement and optimal-barrier evaluations support this search, followed by a capped quadratic program that balances spatial anchoring and projected motion with relative geometry and interval centering. Scalar-profile reconstruction produces the temporal map. Three certified bounds explain unavoidable position error under exact topology, finite widths, and bounded topological error; they benchmark layouts rather than prune the search. TMTM and ST-MTM are exact-topology special cases of the same formulation, so the bounds also measure them. Continuous feasibility and actual rendered errors are reported separately.

Index Terms: Time-varying scalar fields, merge tree maps, spatial anchoring, topology preservation, constrained layout.

1 Introduction

2D and 3D time-varying scalar fields describe evolving phenomena such as pressure systems, combustion structures, and flow features. Their analysis requires understanding where features occur, how they move along a relevant spatial direction, and how their hierarchical relationships change throughout a sequence. Extrema and component merges describe this evolving scalar structure, while spatial coordinates connect it to the physical domain. A temporal overview should bring these aspects together. Figure 1 illustrates how spatial supports and reference positions connect to scalar profiles and the resulting temporal map.

Animation preserves the spatial view at each instant, but comparison between distant times requires navigation and memory. Small multiples expose observations simultaneously at the cost of screen space. Static temporal summaries allocate one display axis to time and compress the spatial domain into the remaining axis [1,31,34]. Spatial linearization makes long sequences accessible in one image; the meaning of its 1D coordinate determines which spatial questions the overview can support.

Merge tree maps organize this compression around scalar features and their hierarchy. Temporal Merge Tree Maps (TMTM) linearize augmented merge trees into scalar profiles aligned across time [22]. Minima and intervening barriers encode extrema and merge levels, exposing scalar structure and its temporal evolution. Spatiotemporal Merge Tree Maps (ST-MTM) incorporate relative spatial distances and temporal position constraints into a geometry-aware layout [9]. These methods provide the foundation for compact overviews of temporal scalar topology.

A display coordinate chosen for hierarchy, relative distance, or temporal alignment need not correspond to a fixed physical direction. Its position cannot then be read directly as longitude or distance along a flow axis, and its changes need not represent motion along that direction. Relative geometry supplies spatial relationships but leaves the absolute coordinate origin undetermined. The remaining question is how to give the temporal map a consistent spatial reference while retaining its hierarchical information.

We introduce *Automatic Spatial Anchoring*: estimating one dominant motion direction from matched extrema over the full sequence, assigning projected targets in that shared coordinate system, and placing display anchors nearby as constraints permit. A task-specific physical direction remains an option. The direction estimator is a standard leading-eigenvector calculation; our contribution is its connection to merge-tree layout, explicit topology budgets and position bounds. A reference order can conflict with topology when the projected leaves of different subtrees interleave while exact topology requires contiguous subtree blocks. Finite feature widths introduce further packing constraints. Allowing bounded topological error releases some ordering restrictions and exposes a controlled position–topology trade-off.

AnchorMTM, *Spatially Anchored Merge Tree Maps*, follows one construction: extract and match features, construct a fixed reference, seek a topology-admissible leaf order, optimize anchors within legal intervals, and reconstruct scalar profiles on a shared temporal axis. Exact Mode and Budgeted Mode use the same layout framework with different admissible merge errors. Position bounds then explain the discrepancy that topology and finite widths make unavoidable.

Our main contributions are as follows:

- We formulate automatic spatial anchoring for temporal merge tree maps, using a single direction estimated from matched extremum motion to establish consistent spatial references across a 2D or 3D sequence.
- We develop a topology-budgeted layout framework that combines reference-aware leaf ordering with capped spatiotemporal anchor optimization. Exact and budgeted modes share the same framework, with explicit position and merge-level error accounting, and existing merge tree maps are special cases of its formulation.
- We establish certified position bounds that quantify unavoidable reference displacement under exact topology, finite feature widths, and bounded topological error.

2 Related Work

2.1 Spatially Referenced Temporal Visualization

Static temporal displays expose distant observations simultaneously, while animation retains the spatial view of each instant; their usefulness depends on the analytical task [31, 34]. Generalized space–time cubes describe the transformations connecting such views [1]. In compressed temporal overviews, the spatial coordinate determines which location and motion relationships remain directly interpretable.

Hovmöller diagrams retain a selected physical coordinate and summarize the other dimensions [20]. Geographic orderings and space-filling curves instead traverse the domain [8, 16]. Dynamic Volume Lines and data-driven curves extend spatial traversal to volumetric comparison and scalar-aware ordering [37, 39]. A traversal can retain spatial neighborhoods, but its 1D coordinate is generally different from a fixed projection, and traversal adjacency alone does not enforce the merge hierarchy.

1D projections organize spatiotemporal phenomena [14], and MotionRugs expose collective behavior of moving entities [5]. More generally, an embedding trades original-domain relationships against its display organization [28]. Relative proximity does not determine an absolute coordinate. AnchorMTM assigns fixed projection targets to topological features and quantifies their displacement under hierarchical and interval constraints, combining spatial reference with a temporal topology overview.

2.2 Topology-Based Temporal Maps and Geometry-Aware Layouts

Topology-based visualization organizes scalar structures through connectivity [17]. Landscapes and landscape profiles express hierarchies through terrain or 1D scalar representations [29, 35], with geometry-preserving and temporally coherent extensions [3, 18]. These representations connect scalar values and feature relationships in a compact view.

TMTM emphasizes intact topological features and temporal alignment in static merge-tree maps [22]. ST-MTM incorporates relative spatial geometry and temporal position constraints [9]. These priorities give spatial relationships and coherence a role in the map, while leaving its display coordinate free to differ from a fixed physical direction. AnchorMTM estimates a shared direction from extremum motion, or accepts a physical direction, and couples that reference to topology-budgeted layout and certified position bounds. It reuses the established profile and scalar-filling foundation.

Hierarchy-constrained leaf ordering traditionally minimizes adjacent dissimilarity [2]; target-order reordering studies related tree-consistent permutations [6]. Separation-constrained layouts control geometric overlap and displacement [11,24]. AnchorMTM couples these layout concerns to fixed spatial targets. Its exact-topology point-order bound has a closed form, whereas finite-width candidate ordering remains heuristic.

Merge-tree comparison depends on the descriptors and correspondences used [38]. Labelled cophenetic/interleaving comparisons support fixed-identity merge-level accounting [15, 27], and ordered merge trees connect scalar curves to monotone interleaving distances [4]. We use fixed labels to measure displayed merge error. The optimal-barrier subproblem relates to minimax ultrametric fitting [13], with the additional restriction that one linear order induces every displayed merge.

3 Background

3.1 Augmented Merge Trees and Feature Supports

For a sampled 2D or 3D field $f_t : D \to \mathbb{R}$, the merge tree T_t records sublevel-set connectivity: minima create components and saddles join them as the threshold increases. Negating the field gives the split-tree convention. Critical points are supernodes connected by superarcs; augmentation associates regular vertices with these arcs [9, 22, 32]. Extraction and simplification precede layout.

A leaf represents an extremum at spatial location p_i^t. Its feature support S_i^t consists of that extremum and the regular samples on its incident leaf arc, excluding the parent critical sample. It is not the full component or the complete subtree above that arc. For cross-frame overlap, these supports are sets of stable spatial sample IDs: the same ID must denote the same spatial sample across frames. Fixed-grid inputs preserve this identity through consistent sample indexing. Internal arcs retain further regular samples for scalar filling. Write $\mathscr{D}_t(v)$ for the descendant leaf labels of a node v, and $h_t(v)$ for its scalar level. The lowest common ancestor determines the merge-height matrix:

$$V_{ij}^t = h_t(\mathrm{lca}(i,j)), \quad i \neq j; \qquad V_{ii}^t = f_t(p_i^t). \tag{1}$$

Equal-level merge events form one multifurcation; arbitrary binary refinements do not add strict constraints. All subsequent guarantees refer to the supplied, possibly simplified tree.

3.2 One-Dimensional Merge-Tree Profiles

A 1D merge-tree profile represents extrema at anchors in a selected leaf order. Regular vertices on leaf arcs supply scalar context within leaf intervals; internal arcs and paths to the LCA supply the intervening gaps [9, 22]. In join convention, extrema become profile minima, and the largest profile value between two anchors gives

their displayed merge level. Exact preservation requires each strict subtree to occupy a contiguous leaf block. Arranging these profiles along time yields a static temporal map. This profile model provides the basis for the spatial anchoring formulation below.

4 AnchorMTM

AnchorMTM connects a shared spatial reference to the displayed leaf anchors. We construct the reference and formulate the layout, describe the construction, and then explain its unavoidable position costs through certified bounds. Analysis uses join convention; split-tree scalar signs are restored after reconstruction.

4.1 Spatial Reference and Problem Formulation

Cross-frame Leaf Correspondence. We track merge-tree leaf extrema p_i^t using their leaf-arc supports S_i^t. On a shared fixed grid, overlap-based preprocessing [9] counts common spatial samples:

$$O_{ij}^t = |S_i^t \cap S_j^{t+1}|. \tag{2}$$

Hungarian assignment maximizes total intersection count over one-to-one pairs, discarding zero-overlap pairs. The objective is not intersection over union (IoU). Figure 2 illustrates the resulting identity states.

Continuation inherits a Track ID; *birth/death* starts an unmatched current identity or ends an unmatched previous track. For *non-bijective correspondence*, including potential one-to-many, many-to-one and ambiguous relations, only the assigned one-to-one links are retained. These identity states neither identify physical split/merge events nor form a branching lineage. For a current leaf i, $\mu_t(i)$ denotes its valid predecessor. The layout receives precomputed tracks aligned to leaf rows; upstream support and confidence conventions are detailed in Supplementary Material.

These correspondences, computed before any topology-budget sweep, provide the motion samples for Spatial Anchoring, which distinguishes a reference target q_i^t from the display anchor x_i^t. Only valid consecutive-frame matches contribute displacements; failed matches or interrupted tracks contribute no sample for that step. By default, one unit direction $a^* \in \mathbb{R}^d$, $d \in \{2,3\}$, is estimated offline from all matched consecutive-frame extremum displacements:

$$\begin{aligned} \Delta p_i^t &= p_i^t - p_{\mu_t(i)}^{t-1}, \\ S &= \sum_{t,i \text{ matched}} \Delta p_i^t (\Delta p_i^t)^\top, \\ a^* &\in \arg\max_{\|a\|=1} a^\top S a. \end{aligned} \tag{3}$$

The leading eigenvector maximizes squared projected motion, not feature count or topological compatibility. The same calculation serves 2D and 3D; it is performed once for the whole sequence. Its sign makes total signed projected motion nonnegative; for near-zero signed motion, the largest-magnitude component is made positive. With negligible motion energy, the longest domain axis is used (first axis on a tie). Repeated leading eigenvalues leave the maximizing direction nonunique; the reported eigengap indicates this ambiguity. A supplied physical direction may replace a^* without changing the layout. Overlap matching may miss fast-moving features or those with changing or weakly overlapping supports, affecting the displacement sample and hence the estimated direction.

For the selected direction, the minimum and span of the projected domain bounding-box corners define a common origin o and extent $L > 0$. The reference is

$$q_i^t = \langle a^*, p_i^t \rangle - o \in [0, L]. \tag{4}$$

Direction, origin and scale remain fixed across frames and all budgets for the same prepared sequence. Implementation arrays store

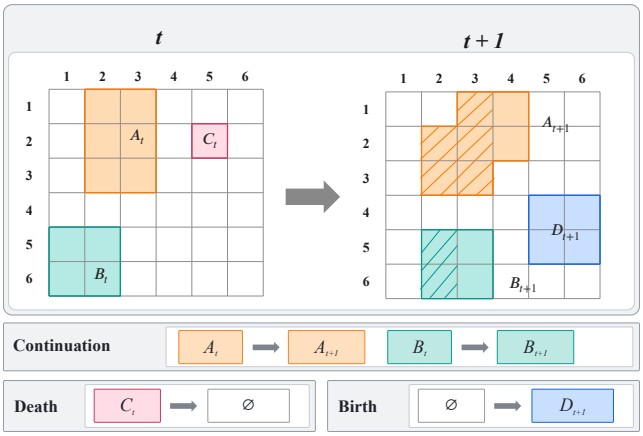


Figure 2: Overlap-based leaf correspondence on a shared grid. Colored regions represent leaf-arc supports; hatching marks samples shared with the previous frame. Maximum-overlap one-to-one assignment continues A and B, leaves C_t unmatched, and assigns a new identity to D_{t+1}. Birth and death denote matching states, not identified physical events.

raw projections on $[o, o+L]$; translating both references and layout coordinates by $-o$ gives the paper convention. The origin is subtracted once, not again by the solver. Distinct extrema can share projected coordinates while retaining separate identities. Anchoring minimizes constrained deviation rather than requiring $x_i^t = q_i^t$.

Relative geometry uses distances d_{ij}^t between leaf-support centroids. These differ from extremum references because a changing support can move its centroid while the extremum stays fixed. The automatic-reference adapter supplies extrema explicitly; the low-level solver also accepts precomputed references and otherwise defaults to centroid projection. Widths $W_i^t = s m_i^t$ convert positive support measures with a fixed scale s, without per-frame normalization.

With correspondence and the reference fixed, a frame comprises a leaf order, extremum anchors, and a scalar profile:

$$M_t = (\pi_t, x_t, \psi_t), \qquad \psi_t : [0, L] \to \mathbb{R}. \tag{5}$$

We seek reference-aligned positions and motion while approximating relative geometry and controlling displayed topology. The interval-layout set $\mathscr{L}(\pi)$ requires intervals of widths W_i centered at z_i to lie inside the canvas, remain disjoint with gap at least $g \geq 0$, and contain anchors satisfying $|x_i - z_i| \leq \rho W_i/2$, $0 \leq \rho \leq 1$. Infeasible total widths and gaps are reported rather than rescaled.

Actual displacement is $|x_i - q_i|$, normalized by L when reported. To distinguish order conflict from finite-width placement, define

$$\mathrm{NROE}(\pi) = \frac{1}{2L} \max_{j \leq k} [q_{\pi(j)} - q_{\pi(k)}]^+. \tag{6}$$

This is the smallest worst displacement of a nondecreasing point placement, divided by L; it ignores widths. For finite intervals, the exact fixed-order cost is

$$\tau(\pi) = \min_{(x,z) \in \mathscr{L}(\pi)} \|x - q\|_\infty. \tag{7}$$

The minimum is attained whenever the layout set is nonempty. It is evaluated exactly from the fixed-order packing constraints, equivalently through a minimax linear program.

For distinct labels, let V_{ij}^{ψ} be the maximum profile value between

their anchors, with $V_{ii}^{\psi} = \psi(x_i)$. We require

$$d_{\mathrm{top}}(\psi_t, T_t) = \max_{i,j} |V_{ij}^{\psi_t} - V_{ij}^{t}| \le \eta, \qquad \eta = (\varepsilon_0 + \varepsilon)\Delta_F. \tag{8}$$

Here $\Delta_F > 0$ is the sequence-wide scalar range, ε_0 is a fixed numerical allowance, and $\varepsilon \ge 0$ is the sole primary user control. The implementation defaults to $\varepsilon_0 = 10^{-7}$. There is no 20% cap: 100% is neither a mathematical upper bound nor necessarily unrestricted topology. NMLE reports the achieved normalized error $d_{\mathrm{top}}/\Delta_F$. Exact topology means zero labelled error on the declared tree. A nonzero tie allowance is accounted for in η.

All remaining settings belong to a fixed configuration Θ_0: position slack β_0, the four objective weights, width scale, gap, anchor slack and output resolution. For a prepared sequence, the default interface is $M_t^{\varepsilon} = \mathrm{AnchorMTM}(T_t, q_t, W_t, V_t, \mu_t; \varepsilon, \Theta_0)$. Advanced settings may be supplied once, but only ε varies in a budget sweep. Supplementary Material specifies the fixed scale rules and implementation tolerances.

For canonical cophenetic merge-tree representations whose fixed labels cover every leaf and whose induced matrices include the diagonal, d_{top} equals the unnormalized labelled interleaving distance [15, 27]. Under these conditions, $d_B \le d_I \le d_{\mathrm{top}}$ [26]. Unlabelled extrema introduced by rendering must be checked before applying this interpretation to the whole displayed tree.

4.2 Spatially Anchored Layout Construction

With features and their reference fixed, construction chooses an order by the lexicographic objective

$$\pi_t^{\varepsilon} \in \arg\min_{\pi \in \mathscr{C}_t(\varepsilon)} \big(\tau_t(\pi), \mathrm{NROE}_t(\pi)\big), \quad \text{s.t.} \quad \delta_t^*(\pi) \le \eta(\varepsilon). \tag{9}$$

The finite candidate pool contains the original strict-topology order, midpoint initializations, and orders visited during width-aware refinement of threshold and selective flattenings. Every candidate is evaluated against the same source merge matrix. Selection minimizes τ first and NROE second; it is not a global optimum over all permutations. Algorithm 1 separates this discrete selection from fixed-order placement.

Algorithm 1: Single-Control AnchorMTM

Data: 2D/3D field sequence and grid; $\varepsilon \ge 0$; fixed configuration Θ_0.
Result: Temporal map I^{ε}, error records, or a failure report.
Extract augmented trees, leaf supports and extrema; match supports and assign Track IDs
Estimate one direction and fixed references q_t; reuse this preprocessing across budgets
$\eta \leftarrow (\varepsilon_0 + \varepsilon)\Delta_F$
for *each time step t in temporal order* **do**
  Report and stop if widths and gaps exceed the fixed canvas
  $\mathscr{C}_t \leftarrow \mathrm{GenerateAndRefine}(T_t, q_t, W_t, V_t, \eta)$; retain strict-topology candidates
  Evaluate $\tau(\pi)$, NROE(π) and $\delta^*(\pi)$ for candidates
  $\mathscr{F}_t \leftarrow \{\pi \in \mathscr{C}_t : \delta^*(\pi) \le \eta\}$
  if $\mathscr{F}_t = \varnothing$ **then** report no feasible candidate and stop
  Select π_t^{ε} from $\mathscr{F}_t$ by (τ, NROE)
  Compute optimal merge barriers for π_t^{ε}
  Solve the four-term QP only for π_t^{ε}, with cap $\tau(\pi_t^{\varepsilon}) + \beta_0$ and valid matched history
  Reconstruct the profile by leaf/LCA filling and barrier correction
  Record candidate counts, continuous and rendered errors, motion and failures; append profile to I^{ε}
return I^{ε} *and diagnostic records*

4.2.1 Topology-Budgeted Leaf Ordering

Exact Mode sets $\eta = 0$ and requires zero labelled merge error on the supplied tree. *Budgeted Mode* permits $\eta > 0$. Setting only the additional budget $\varepsilon = 0$ while retaining $\varepsilon_0 > 0$ gives a numerical allowance, not zero labelled error. Both modes share the same interval model and placement objective.

Candidate generation includes the exact tree, threshold flattenings, and a conflict-driven selective flattening. Midpoint orders initialize these structures. Width-aware block insertion refines the four best initial structures, together with the exact tree, while retaining the point-model budget order as a candidate. Moves improve the lexicographic score consisting of τ and a squared packing-residual term, and must pass $\delta^*(\pi) \le \eta$. Visited initial and trial orders are retained in the deduplicated pool, including the strict-tree initialization and retained point candidate. Final budget filtering and lexicographic selection use the exact fixed-order costs; refinement does not certify a global optimum over permutations.

Guards report capacity failures, empty admitted sets and solver failures instead of silently rescaling widths, gaps or canvas; the supplement details them.

4.2.2 Optimal Merge Barriers and Topological Feasibility

For a fixed order, a pair's displayed merge is the largest intervening barrier. Set $C_k = \min_{i \le k < j} V_{\pi(i)\pi(j)}$ and $F_k = \max(V_{\pi(k)\pi(k)}, V_{\pi(k+1)\pi(k+1)})$ for gap k.

Proposition 1 (Optimal barriers)**.** *With fixed extremum values in the continuous non-strict model, the smallest labelled merge error is*

$$\delta^*(\pi) = \max\left\{0, \max_k (F_k - C_k), \frac{1}{2}\max_{i<j}\left(V_{\pi(i)\pi(j)} - \max_{i \le k < j} C_k\right)\right\}, \tag{10}$$

and barriers $b_k = C_k + \delta^*(\pi)$ attain it.

Error δ requires $F_k \le b_k \le C_k + \delta$. Choosing the largest admissible barriers satisfies every pair's lower merge requirement exactly when $\delta \ge \delta^*$. Thus the order meets the merge-level allowance in this continuous model exactly when $\delta^*(\pi) \le \eta$. This fixed-order test supports candidate search without making it globally optimal. Strict-extremum clearance and raster effects are checked separately.

4.2.3 Spatially Anchored Spatiotemporal Placement

Once the order is fixed, the core terms are E_R, mean squared deviation from the fixed reference, and E_M, mean squared error in matched projected motion. The auxiliary term E_G approximates support-centroid distances through ordered anchor separations; E_C regularizes anchor offsets from interval centers. All four terms and their nonnegative weights are retained:

$$\min_{x_t, z_t} \ \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \quad \text{s.t.} \quad (x_t, z_t) \in \mathscr{L}(\pi_t), \ \|x_t - q_t\|_\infty \le \tau(\pi_t) + \beta_0. \tag{11}$$

For each valid match, set $\Delta q_i^t = q_i^t - q_{\mu_t(i)}^{t-1}$ and use the previous display anchor plus this reference displacement as the temporal target:

$$\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + (q_i^t - q_{\mu_t(i)}^{t-1}). \tag{12}$$

Thus E_M follows projected motion while favoring continuity of $x - q$. First-frame, newly born and otherwise unmatched leaves have no previous-frame motion penalty. Each term is averaged over

its features, ordered pairs or valid matches; the current automatic-reference adapter uses unit confidence for valid matches. Supplementary Material distinguishes this default from legacy confidence-weighted paths.

Only ε changes the topology allowance $\eta(\varepsilon)$. The fixed slack $\beta_0 \geq 0$ bounds additional position deviation after ordering. The cap is feasible because $\tau(\pi_t)$ is attained; the QP receives only the selected order and cannot replace it by one with a lower quadratic objective. It bounds continuous displacement while geometry and temporal objectives select a compromise within that cap. The fixed-order problem is convex; sequential processing does not imply a joint sequence optimum or exact preservation of every projected motion.

4.2.4 Profile Reconstruction and Output Diagnostics

Reconstruction inherits the leaf-arc sampling and LCA-path filling of the merge tree map pipeline [9, 22]. Original extrema remain at their anchors; regular samples provide scalar context within intervals and gaps. Optimal-barrier correction then clips each inter-anchor interior to b_k and raises an existing peak if necessary. This separate correction is required for relaxed orders; inherited LCA filling alone does not enforce the minimax barriers.

A common direction, origin, extent, and pixel conversion are used for all frames. The profiles form the final temporal map,

$$I^{\varepsilon}(t,u) = \psi_t^{\varepsilon}(u), \qquad 0 \leq u \leq L. \tag{13}$$

There is no per-frame recentering. Resampling does not retain every source sample, and strict-extremum clearance and raster rounding can change the achieved error; reconstruction failures and full-tree label coverage are checked as described in the supplement. A recorded profile is therefore not unconditionally certified by $\delta^* \leq \eta$. Records distinguish the requested budget, position certificates, attained τ, continuous δ^*, and actual rendered errors.

4.3 Certified Position–Topology Bounds

The construction can leave reference displacement even after optimization. The following bounds explain what topology and finite widths make unavoidable: the first answers how much a tree-compatible point order must deviate, the second adds interval packing, and the third asks which restrictions survive a merge-error allowance. They are analytical benchmarks, not three execution stages or search-pruning rules; construction uses the exact fixed-order costs τ and δ^*. The resulting certificates apply independently of the layout algorithm. Full proofs and computational definitions accompany the paper in Supplementary Material. We suppress t within a frame, use $[v]^+ = \max(v,0)$, and take empty maxima as zero.

4.3.1 Exact-Topology Reference-Order Bound

For a reduced tree with strictly increasing proper-ancestor levels, exact topology makes every subtree contiguous. Let $[m_X, M_X]$ be the reference hull of the leaves in subtree X.

Theorem 1 (Exact-Topology Position Bound)**.** *Every zero-error profile has* $\mathrm{NROE}(\pi) \geq \nu^*(T,q)$*, where*

$$\nu^*(T,q) = \frac{1}{2L} \max_{v} \max_{\substack{X,Y \in \mathrm{ch}(v) \\ X \neq Y}} [\min(M_X - m_Y, M_Y - m_X)]^+. \tag{14}$$

Recursive child ordering by reference-hull midpoint attains this point-model order bound in $O(n \log n)$ time.

Proof sketch. Sibling blocks must choose one orientation. Their two possible inversion costs are $[M_X - m_Y]^+/2$ and $[M_Y - m_X]^+/2$. Midpoint order chooses the smaller; every inverted leaf pair is separated by sibling blocks at its LCA. □

This result isolates the ordering restriction imposed by the tree and references. Midpoint ordering attains the point-order optimum; finite intervals introduce additional constraints, considered next.

4.3.2 Width-Aware Position Bound

Finite widths, minimum gaps, anchor slack and canvas capacity add packing costs to the point-order obstruction. For a nonempty leaf set S, let $H_i = \rho W_i/2$, $\ell_i = q_i - W_i/2 - H_i$, $h_i = q_i + W_i/2 + H_i$, and $s_i = W_i + g$. Define two packing summaries:

$$\begin{aligned} A(S) &= \max_{a \in S} \left(\ell_a + \sum_{\substack{i \in S \\ \ell_i \geq \ell_a}} s_i \right), \\ B(S) &= \max_{b \in S} \left(\sum_{\substack{i \in S \\ h_i \leq h_b}} s_i - h_b \right). \end{aligned} \tag{15}$$

The first uses upper reference-endpoint sets and the second lower sets; sorting and cumulative sums evaluate them. For disjoint leaf sets X, Y and the full leaf set $\mathscr{F}$, write

$$\begin{aligned} \omega(X,Y) &= \tfrac{1}{2} \min\{A(X)+B(Y), A(Y)+B(X)\} - \tfrac{g}{2}, \\ E_{\partial} &= \max\{0, B(\mathscr{F}) - g, A(\mathscr{F}) - g - L\}. \end{aligned} \tag{16}$$

Here ω takes the cheaper block orientation and E_{∂} accounts for both canvas ends. Let $\mathscr{P}_0$ contain every pair of distinct sibling descendant-leaf sets. The certificate is

$$\mathscr{B}_w(T,q,W) = \max\{E_{\partial}, \max_{(X,Y) \in \mathscr{P}_0} \omega(X,Y)\}. \tag{17}$$

The dependence on g, ρ and L is through Eqs. 15–16; it is suppressed in the certificate's arguments.

Theorem 2 (Width-Aware Position Bound)**.** *For any feasible exact-topology interval layout,*

$$\|x - q\|_{\infty} \geq \tau(\pi) \geq \mathscr{B}_w(T,q,W). \tag{18}$$

With zero widths and gaps, $\mathscr{B}_w = L\nu^*(T,q)$.

Proof sketch. The first and last anchors of selected leaves must enclose their intervals and intervening gaps, up to the allowed anchor slack. Upper and lower sets in Eq. 15 maximize the resulting shortfall against reference separation. Exact topology requires one orientation of every sibling pair; the canvas ends give the two one-sided costs. □

Thus $\mathscr{B}_w$ bounds all exact-topology legal interval orders and need not be attained; $\tau(\pi)$ is the attained minimum for a specified order. These quantities have coordinate units, whereas NROE and ν^* are normalized by L.

4.3.3 Bounded-Topology Position Bound

A merge-level allowance can release some contiguous blocks, but still limits interleaving. If leaf k lies between descendants i, j of node u, interval containment gives $V_{ik}^{\psi} \leq V_{ij}^{\psi}$, hence $V_{ik} \leq h(u) + 2\eta$. Define $A_\eta(u)$ as the highest ancestor, including u, with $h(A_\eta(u)) - h(u) \leq 2\eta$. This uses the accumulated ancestor-height difference in scalar units, on the reduced equal-level tree with the fixed labels of Eq. 8.

Let $\mathscr{P}_\eta$ consist of two types of leaf-set pairs: $(\mathscr{D}(u), \{k\})$ for every internal node u and leaf $k \notin \mathscr{D}(A_\eta(u))$; and $(\mathscr{D}(u), \mathscr{D}(w))$ for internal nodes u, w satisfying

$$\begin{aligned} \mathscr{D}(w) \cap \mathscr{D}(A_\eta(u)) &= \varnothing, \\ \mathscr{D}(u) \cap \mathscr{D}(A_\eta(w)) &= \varnothing. \end{aligned} \tag{19}$$

For any nonempty leaf set X, use its reference hull $[m_X, M_X]$ and define $\kappa(X,Y) = [\min(M_X - m_Y, M_Y - m_X)]^+/(2L)$. The two certificates are

$$C_\eta(T,q) = \max_{(X,Y)\in\mathscr{P}_\eta} \kappa(X,Y),$$
$$C_{w,\eta}(T,q,W) = \max\{E_\partial, \max_{(X,Y)\in\mathscr{P}_\eta} \omega(X,Y)\}. \tag{20}$$

Theorem 3 (Bounded-Topology Position Bound)**.** *In a profile with $d_{\text{top}} \le \eta$, no leaf outside $\mathscr{D}(A_\eta(u))$ can lie between two leaves of u. Every pair in $\mathscr{P}_\eta$ lies wholly in one block orientation or the other. Consequently,*

$$\mathrm{NROE}(\pi) \ge C_\eta(T,q), \qquad \tau(\pi) \ge C_{w,\eta}(T,q,W), \tag{21}$$

with the second inequality applying to the interval model.

The separation principle excludes interleaving for each eligible pair; the two possible orientations then yield κ and ω. Thus C_η bounds normalized reference-order error, while $C_{w,\eta}$ includes finite-width and boundary costs in coordinate units. Accumulated ancestor gaps make these certificates at least as strong as those obtained by flattening every parent gap at most 2η.

The allowance concerns the displayed hierarchy; the source field and feature identities stay fixed. A larger feasible set need not improve every feature or every heuristic result. These bounds are reported as benchmarks for unavoidable error; ranking and continuous acceptance use the exact fixed-order evaluations in Section 4.2. Continuous feasibility does not certify the full rendered raster.

4.4 Existing Merge Tree Maps as Special Cases

The construction above is one instance of a more general formulation. Every merge tree map we consider chooses, per frame, an order, interval layout and profile that solve

$$\begin{aligned} \min_{\{M_t\}} \quad & \textstyle\sum_t \lambda_O E_O + \lambda_G E_G + \lambda_M E_M + \lambda_R E_R + \lambda_C E_C + \lambda_W E_W \\ \text{s.t.} \quad & \pi_t \in \Pi_\eta(T_t), \quad (x_t, z_t) \in \mathscr{L}(\pi_t), \quad \|x_t - q_t\|_\infty \le B_t, \end{aligned} \tag{22}$$

where $\Pi_\eta(T_t)$ contains the orders whose optimal barriers meet the allowance, $\delta_t^*(\pi) \le \eta$, E_O is the subtree-overlap term of TMTM, E_W the width-deviation term of ST-MTM, and E_G, E_M, E_R, E_C are the terms of Eq. 11. The allowance η, the position cap B_t, the rule that selects π_t and the objective weights determine a map (Table 1). TMTM and ST-MTM are exact-topology instances with no position cap and no reference term; the projection is the instance with unlimited allowance, whose order is the reference order; AnchorMTM fixes the reference term and the cap, and exposes the allowance as its single control.

The bounds of Section 4.3 depend only on the constraints of Eq. 22, not on the objective or the solver, so they measure every instance to which their assumptions apply. Theorem 1 holds for every exact-topology map, whatever its width rule or solver, and hence for TMTM and ST-MTM. Theorem 2 applies to exact-topology maps in the interval model $\mathscr{L}$, which ST-MTM and AnchorMTM share, and Theorem 3 to every map whose merge error stays within η. For any exact-topology map, the order error above ν^* is therefore a consequence of its order rule rather than of the data, and Section 5 measures it.

5 Experiments

The experiments test the three claims of the framework: anchoring removes the avoidable order error of exact maps (Section 5.2), the bounds identify what remains (Section 5.3), and ε traces the position–topology trade-off within its allowance (Section 5.4); Sections 5.5 and 5.6 add case studies, the ordering and its cost.

Table 1: Merge tree maps as instances of Eq. 22.

Map	allowance η	cap B_t	order rule	objective
TMTM [22]	0	∞	greedy subtree flips	E_O
ST-MTM [9]	0	∞	spatial proximity	$E_G + \lambda_M E_M$; E_W
Projection	∞	$\tau_t + \beta_0$	sort by q	Eq. 11
AnchorMTM	$(\varepsilon_0 + \varepsilon)\Delta_F$	$\tau_t + \beta_0$	Algorithm 1	Eq. 11

5.1 Experimental Setup

Data. The benchmark comprises seven sequences and 917 frames (Table 2): reanalysis pressure, satellite radiation, fire emission and air quality on planar domains, two volumetric flow simulations, and the synthetic Ring sequence of TMTM [7,10,14,19,22,23,30,36]. All maps share the processed fields, augmented trees, leaf supports, correspondences and references. References follow Eq. 4; on Tangaroa the direction is the estimated motion direction (Eq. 3), while on Storms and Cylinder we supply the physically meaningful west–east and streamwise axes. Two further sequences illustrate the maps and enter none of the benchmark aggregates: the 744 hourly Storms fields of Figure 1 and a 90-frame volumetric Vortex sequence.

Instances. We evaluate the four instances of Table 1. AnchorMTM runs in Exact Mode ($\varepsilon = 0$, with the numerical tie allowance $\varepsilon_0 = 10^{-7}$) and in Budgeted Mode ($\varepsilon = 10\%$ unless stated) with its default configuration Θ_0. TMTM and ST-MTM are evaluated on their native outputs; ST-MTM has no public code, and we use our implementation of its published description. Given their native leaf orders, our implementation of Eq. 22 reproduces both: the TMTM linearization matches the official implementation in all 3127 arc intervals and anchors, and the ST-MTM anchors satisfy the optimality conditions of its quadratic programs (relative KKT residuals below 10^{-6}). Both order rules are heuristic even for their own objectives: ST-MTM's order misses its exhaustive optimum on 10 frames, and TMTM's greedy choice is up to 85.5% above the per-step optimum.

Measures. Position is measured by NROE (Eq. 6), which compares maps with different native scales by their order alone, and, for maps in the interval model, by the pooled mean E_{pool} and the worst E_∞ of $|x_i - q_i|/L$ over all feature–frames. Topology is measured by NMLE, the rendered d_{top}/Δ_F (Eq. 8). The motion error (ME) is the fraction of clear approach or separation events, i.e., changes of the 2D or 3D distance between two matched extrema over two steps, that a map shows in the wrong direction or not at all. To credit the objective of ST-MTM, we also report scale-normalized stress (SNS) and trustworthiness (TW, $k = 1$) of anchor distances against support-centroid distances with its protocol [9]; recomputed from the saved native anchors, they reproduce the stored baseline values to a relative 10^{-5} and the third decimal.

5.2 Spatial Anchoring Removes the Avoidable Order Error

Most of the order error of existing merge tree maps is avoidable, and Exact Mode removes it without giving up the merge tree. Because TMTM and ST-MTM are exact-topology instances, Theorem 1 splits their order error into the part ν^* that the hierarchy forces and an excess that their order rule chooses (Fig. 3). Across the seven sequences, the excess accounts for 54–95% of TMTM's order error and 57–95% of ST-MTM's; on Storms, for example, TMTM's 19.2% contains 3.8% that the hierarchy forces and ST-MTM's 16.3% contains 12.5% that it does not. Exact Mode lies within 0.03 percentage points of ν^* on every planar sequence, with zero merge error. On Tangaroa, the width-aware search accepts a slightly higher NROE (9.67% against 9.24%) to lower the worst actual displacement (Section 5.6).

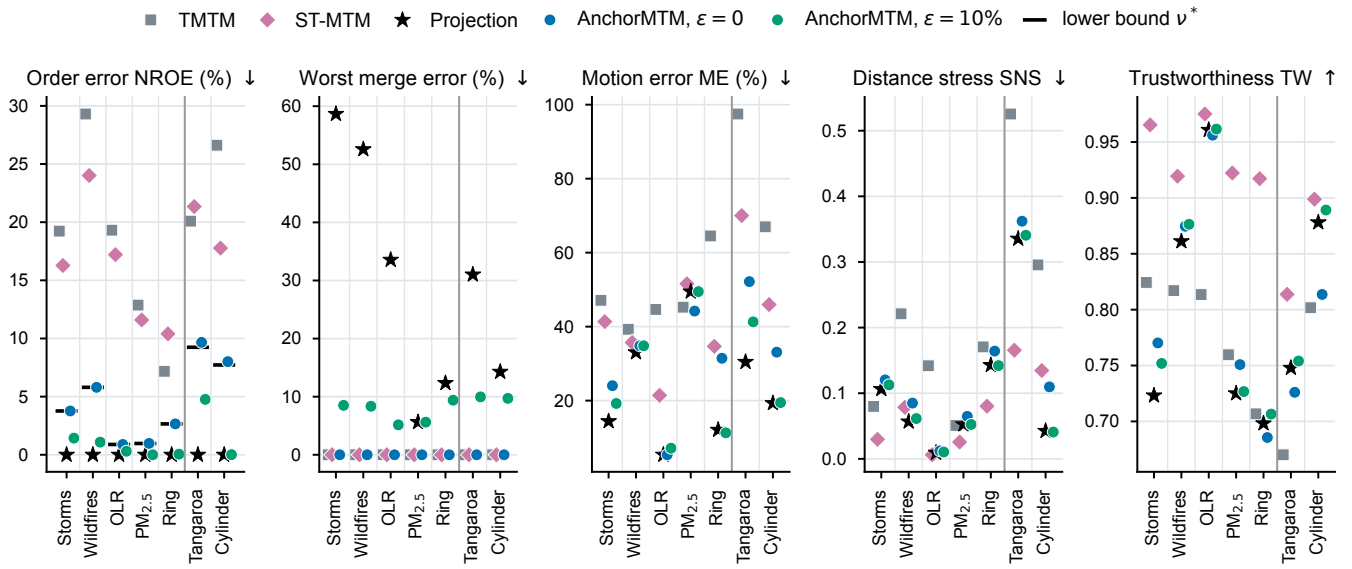


Figure 3: Instances of Eq. 22 on seven sequences (left of the grey line: planar; right: volumetric). Order error NROE with the exact-topology bound ν^* of Theorem 1 (black ticks), worst rendered merge error, motion error, and the distance measures SNS and TW. Exact-topology instances (TMTM, ST-MTM, AnchorMTM at $\varepsilon = 0$) have zero merge error; the projection has zero order error but up to 59% merge error. All values are listed in the supplement.

Table 2: Benchmark sequences. Leaves: retained extrema per frame.

Sequence	Scalar field	Frames	Leaves	Tree
Storms	ERA5 mean sea-level pressure, 2D	118	1–8	join
Wildfires	log(1 + FRP), 2D	30	6–11	split
OLR	outgoing longwave radiation, 2D	60	2–8	join
$PM_{2.5}$	$PM_{2.5}$ concentration, 2D	366	1–7	split
Ring	synthetic ring intensity, 2D	40	1–7	split
Tangaroa	velocity magnitude, 3D	201	18–93	join
Cylinder	Q criterion, 3D	102	4–75	split

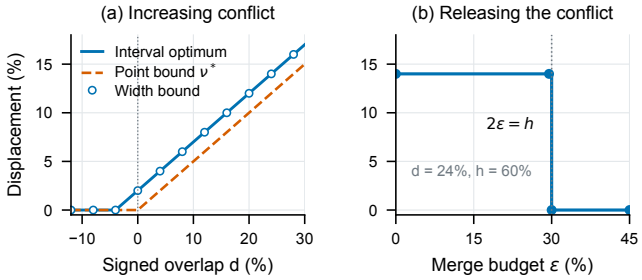


Figure 4: Controlled four-leaf trees. (a) Increasing sibling-hull overlap forces displacement; circles mark the width bound of Theorem 2, which the exhaustive interval optimum attains. (b) At overlap $d = 24\%$, the 14% interval displacement disappears only when the merge budget reaches half the parent gap $h = 60\%$, as Theorem 3 predicts. Percentages refer to a unit canvas or a unit scalar range.

The anchored axis carries motion. A feature that moves along the reference now also moves on the map, so Exact Mode makes fewer motion errors than TMTM and ST-MTM on all seven sequences, for example 5.4% against 21.4% on OLR and 52.2% against 70.0% on Tangaroa. The projection, which follows the reference order exactly, is lower still on most sequences, but only by breaking the tree: it changes merge levels by up to 58.66% of the scalar range on Storms, 52.58% on Wildfires, 33.56% on OLR and 31.02% on Tangaroa, where no frame keeps its tree. Budgeted Mode closes part of this gap within its allowance (Section 5.4); no rendered AnchorMTM map exceeded its declared allowance beyond numerical tolerance.

The price of a spatial axis is stated, not hidden. Anchoring to one direction trades two-dimensional distance preservation for position along that direction. ST-MTM, whose objective is pairwise distance, has the best TW on all seven sequences and the best SNS on five. Where widths dominate, the trade-off can reverse: on Cylinder, Budgeted Mode reaches an SNS of 0.041 against ST-MTM's 0.135. In the maps of the supplement, Exact Mode places the OLR low-radiation basins near their references, so the map is comparable to a time–longitude plot that keeps the nesting of the basins.

5.3 The Certified Bounds Identify the Remaining Displacement

Each bound isolates one cause of displacement. To separate the causes, we fix two sibling pairs $X = \{A, B\}$ and $Y = \{C, D\}$ that merge at 0.4 below a root at 1, with leaf values 0 and references $(0.2, 0.5, 0.5 - d, 0.8 - d)$ on a unit canvas, and vary the signed overlap d from -0.12 to 0.30. Across 43 overlaps and four widths, we compare the bounds with all 24 leaf orders, each placed by an independent minimax LP. Topology alone forces exactly $\nu^* = \max(0, d)/2$. With widths $W = 0.06$, gap 0.01 and $\rho = 0.5$, the width bound, our search and the exhaustive optimum coincide at $\tau = \max(0, (d + 0.04)/2)$ (Fig. 4a): 12% and 14% at $d = 0.24$. A merge allowance removes the conflict exactly where Theorem 3 admits interleaving, at $2\varepsilon = h$ for all four parent gaps tested (Fig. 4b). These are controlled instances, not a general tightness claim.

On real trees, the bounds remain informative. The width bound is attained on most planar frames and on the worst Tangaroa and Cylinder frames, but not uniformly: on Wildfires at $\varepsilon > 0$ and on one OLR frame at $\varepsilon = 10\%$, a worst-frame gap remains. Accumulating ancestor gaps keeps the bounded-topology bound useful on large trees: on Tangaroa at $\varepsilon = 5\%$, flattening every gap below 2η certifies only 0.38% on average, whereas Theorem 3 with widths

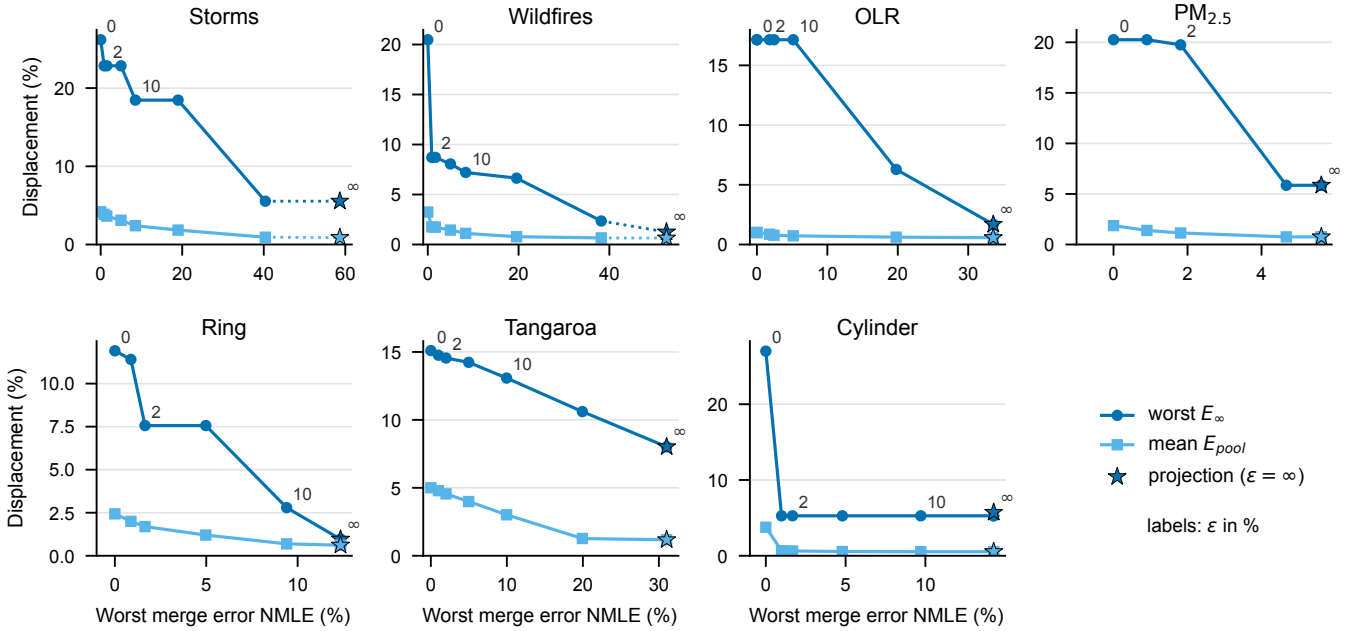


Figure 5: Position–topology trade-off traced by the single control ε. Each point is one map of all frames at budget ε (labels, in percent), placed by its rendered worst merge error and its worst (dark) and mean (light) displacement. Stars: projection ($\varepsilon = \infty$).

certifies 7.40% against an attained 8.10%. A gap of similar size remains at $\varepsilon = 0$, which suggests that the slack lies in the bound rather than in the search. Per frame, the certificates thus tell an analyst how much of a map's displacement no layout in the model could avoid.

Because the bounds need no layout, they inform the choice of reference before a map is drawn. In 3D, a projection discards two coordinates, and the unavoidable order error depends on the direction. We evaluated v^* for 400 sampled directions on both volumetric sequences, with references built as in Eq. 4 (Fig. 6). Over these samples its mean ranges from 4.70% to 10.14% on Tangaroa and from 5.26% to 28.15% on Cylinder. On Tangaroa, the estimated motion direction is among the most conflicting: only 3 of the 400 sampled directions have a higher bound, and the x axis would lower v^* from 9.24% to 6.09%. On Cylinder, the streamwise axis lies near the median (7.72%). In every key direction, Exact Mode stays within 0.6 percentage points of the bound, except on the Tangaroa z axis (2.6 points), where large widths dominate, while TMTM and ST-MTM stay far above it.

5.4 One Control Traces the Position–Topology Trade-off

One control spans the trade-off from exact topology to the reference order, and where the curve bends depends on the source of the conflict. Figure 5 traces each sequence from Exact Mode to the projection; every point respects its allowance. Where conflicts come from packing, small budgets suffice: on Cylinder, $\varepsilon = 1\%$ lowers the worst displacement from 27.00% to 5.27%, below the projection's 5.73%, because sorting by reference ignores widths; on Wildfires, 1% lowers E_∞ from 20.48% to 8.71%; on $PM_{2.5}$, 5% already reaches the projection. Where conflicts come from strong saddles, position is expensive: on Storms, the worst displacement stays above 18% up to $\varepsilon = 20\%$, and on OLR one frame keeps $E_\infty = 17.14\%$ up to 10%. Budgets are spent where they are needed: on Storms at 10%, the mean merge error is 0.90% while the worst frame uses 8.51%.

Aggregate gains are not pointwise gains. In a four-leaf control, enumerating all 24 orders at $\varepsilon = 30\%$ lowers the optimal worst displacement from 22% to 1%, yet one feature's error rises from 0.087% to 1%. On the Vortex diagnostic, raising ε from 20% to 30% more than halves the pooled error (8.79% to 4.27%), while 30.91% of 12,446 feature–frame observations worsen, 24.72% by at least 0.5 percentage points of L. Recorded anchors and references let such local changes be inspected before a budget is accepted.

5.5 Case Studies

Storms: an exact map that reads as geography. On the 118-frame Storms benchmark, the minima are low-pressure systems crossing the North Atlantic, and the reference is the west–east axis. Measured by the worst displacement per step, averaged over steps, the TMTM and ST-MTM objectives place features 37.3% and 50.7% of the canvas from their references within the same layout model, and Exact Mode 5.1% (median 0.8%, peaks of 26.2% in a few steps). A 10% budget lowers the mean to 2.7%; it is spent in 31 of 118 steps, at most 8.51% and 0.90% on average, and the 89 barriers whose merge level changed by more than 0.1% of the scalar range are marked on the map (supplement). Figure 1 applies the construction at $\varepsilon = 5\%$ to 744 hourly fields of December 1999 with a fixed projected direction. Its snapshots follow one matched minimum, a strong storm over Scandinavia, from 16 to 19 December, whose displayed positions form one continuous track through the outlined window; the track is a computational correspondence of leaf features, not by itself evidence of physical storm genesis or dissipation.

Across domains: one construction for planar and volumetric fields. The supplement follows one tracked feature per dataset, on three planar and two volumetric sequences, through 18–41 time steps under six budgets from 0 to 30%, each row keeping the same feature and reference. Read against the reference trajectory, these rows show feature by feature what a larger budget changes, which the aggregate curves of Figure 5 cannot show.

5.6 Ordering and Cost

Width-aware ordering. An order that is optimal for points need not be optimal for intervals when features are large. On Tangaroa, ordering by NROE alone, i.e., the point-model budget order,

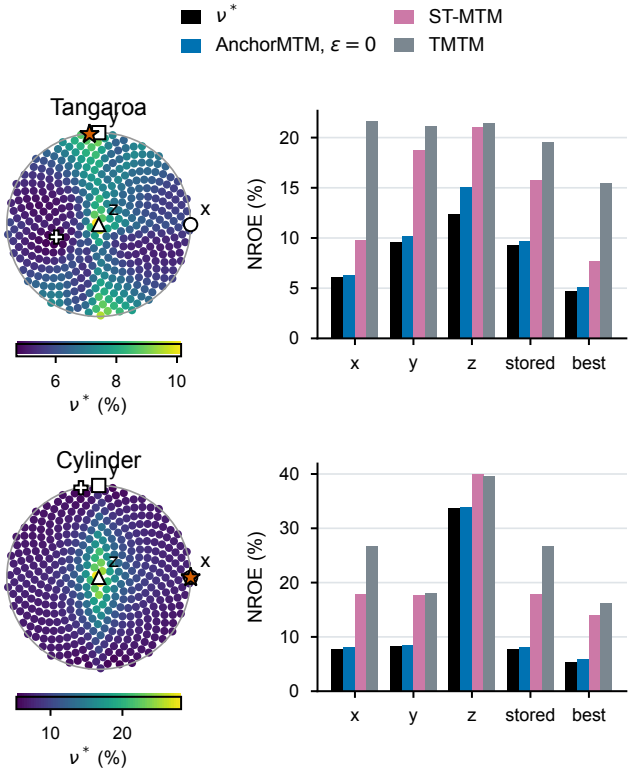


Figure 6: Choosing the reference in 3D. Disks: mean order bound ν^* over 400 sampled directions of the hemisphere (equal-area projection; centre $= z$, rim $= xy$ plane); markers: coordinate axes, the stored direction (star; estimated motion direction on Tangaroa, streamwise axis on Cylinder) and the smallest bound (cross). Bars: ν^* and mean order error of Exact Mode, ST-MTM and TMTM for the key directions (native orientations chosen in their favour).

leaves a worst displacement of 20.69% at $\varepsilon = 0$. Ranking candidates by the exact interval cost τ, as Algorithm 1 does, lowers it to 15.10% at $\varepsilon = 0$ and to 14.55% at $\varepsilon = 2\%$, where the worst frame meets the lower bound, at a mean temporal jump of 2.82% against 2.73%. On Cylinder, $\varepsilon \geq 2\%$ lowers E_∞ from 5.73% with the point-model order to 5.27%, again at the bound.

Search coverage and runtime. On all 637 frames with at most eight leaves, the order selected by the finite search equals the exhaustive optimum at nine budgets from 0 to 100%. The bounds cost $O(n \log n)$ to $O(n^2 \log n)$ and δ^* costs $O(n^3)$ per order. On an Apple M2 Pro (single Python process; NumPy 2.0, SciPy 1.13), all bounds take at most 0.6 ms per planar frame (1–11 leaves) and 5.5 ms per volumetric frame (up to 93 leaves), the ordering at most 3.7 ms and 103 ms, and the capped QP (SciPy SLSQP) 2.3–9.0 ms and 0.25–0.50 s. All 917 frames take 94 s at $\varepsilon = 0$ and at most 137 s with a budget.

6 Discussion and Limitations

One formulation for merge tree maps. Writing merge tree maps as instances of Eq. 22 separates three decisions that existing methods make implicitly: what the display axis means (the reference and its cap), how much topology may change (the allowance), and which secondary geometry to keep (the objective). It also gives every exact map a common yardstick: because midpoint ordering attains ν^*, any excess above it is avoidable by design, while for the width and bounded-topology bounds, which need not be attained, the excess estimates from above what a better layout could remove.

What a budget changes. A budget never removes a feature, changes an extremum value or moves the data. In the continuous model it changes merge levels by at most $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, and every rendered map we measured stayed within this allowance. Where the labels cover all leaves, as we verified on 200 rendered columns of Ring and OLR, the displayed tree and its persistence diagram stay within the measured merge error of the source in interleaving and bottleneck distance. Persistence simplification instead deletes the features below 2ε [12, 33]. Where a subtree is split, its band is no longer contiguous and "one band, one component" fails locally, which is why such barriers are marked.

Using AnchorMTM. The results suggest a simple procedure. First, compare candidate references by ν^*, which takes milliseconds per frame, and choose between the estimated motion direction and a physically meaningful one. Second, start in Exact Mode, whose position error is close to the certified bound on planar data and whose merge tree is exact. Third, sweep ε, choose the smallest budget at which the trade-off curve drops, and inspect the marked barriers and per-feature residuals before accepting it.

Limitations. One direction cannot encode the other coordinates: features with the same projection compete for the same place, and the estimated direction, which follows the dominant motion, need not be the least conflicting one. It is estimated offline from all matches; online use requires a supplied direction or one fitted to an earlier interval. The construction inherits the correspondence: missed or spurious matches change the estimated direction and the motion term. The order search is a finite heuristic; it matched exhaustive enumeration on all frames with up to eight leaves, and on larger trees its excess is limited only by the gap to the bound. Continuous feasibility does not certify the rendered raster: resampling, strict-extremum clearance and pixel rounding can change the achieved merge error, so we report rendered errors separately from δ^*. All results use the default configuration Θ_0; we did not study its sensitivity. Position is measured against the reference that AnchorMTM targets, so we also report the scale-free order error of native outputs and the distance measures of ST-MTM, on which ST-MTM remains better in TW and on most sequences in SNS. Finally, the motion error is a geometric proxy that assumes known identities; whether analysts read position and nesting together, and whether disclosure marks prevent misreading [21, 25], requires a user study.

7 Conclusion

AnchorMTM gives temporal merge tree maps a spatial meaning. A reference, estimated from feature motion or supplied by the analyst, turns the display axis into a spatial coordinate; one control decides how much topology may change to follow it; and three certified bounds state what exact topology, finite widths and a bounded topological error make unavoidable. Because existing merge tree maps are instances of the same formulation, the bounds measure them too: on seven planar and volumetric sequences, 54–95% of their order error is avoidable, and Exact Mode removes nearly all of it on planar data without changing the merge tree. A budget recovers further position where the data allow, with the merge error within its declared allowance and its changes marked on the map. Next steps are scalable order search for large trees, references beyond a single direction, and a user study of reading position and merge levels together.

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